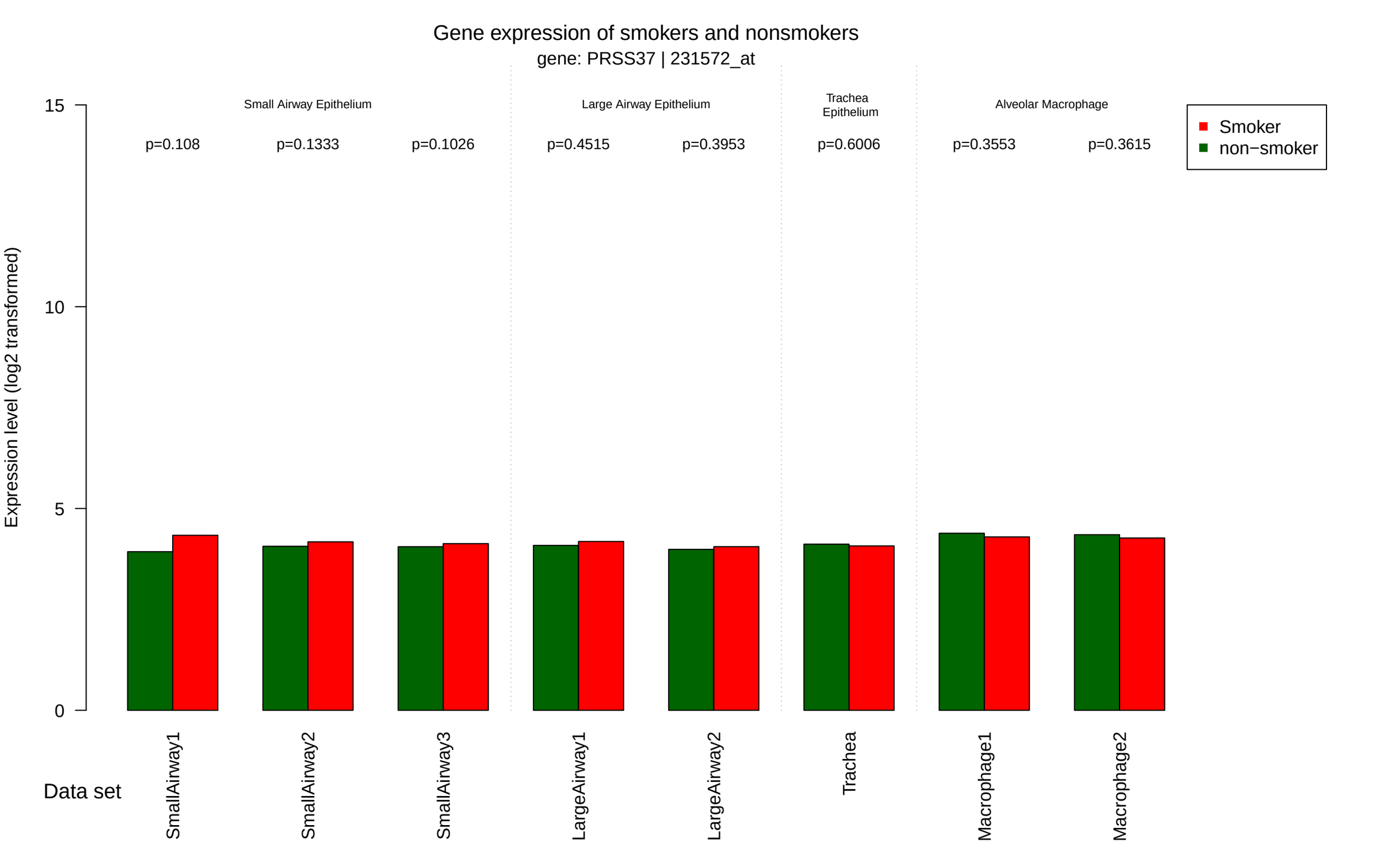
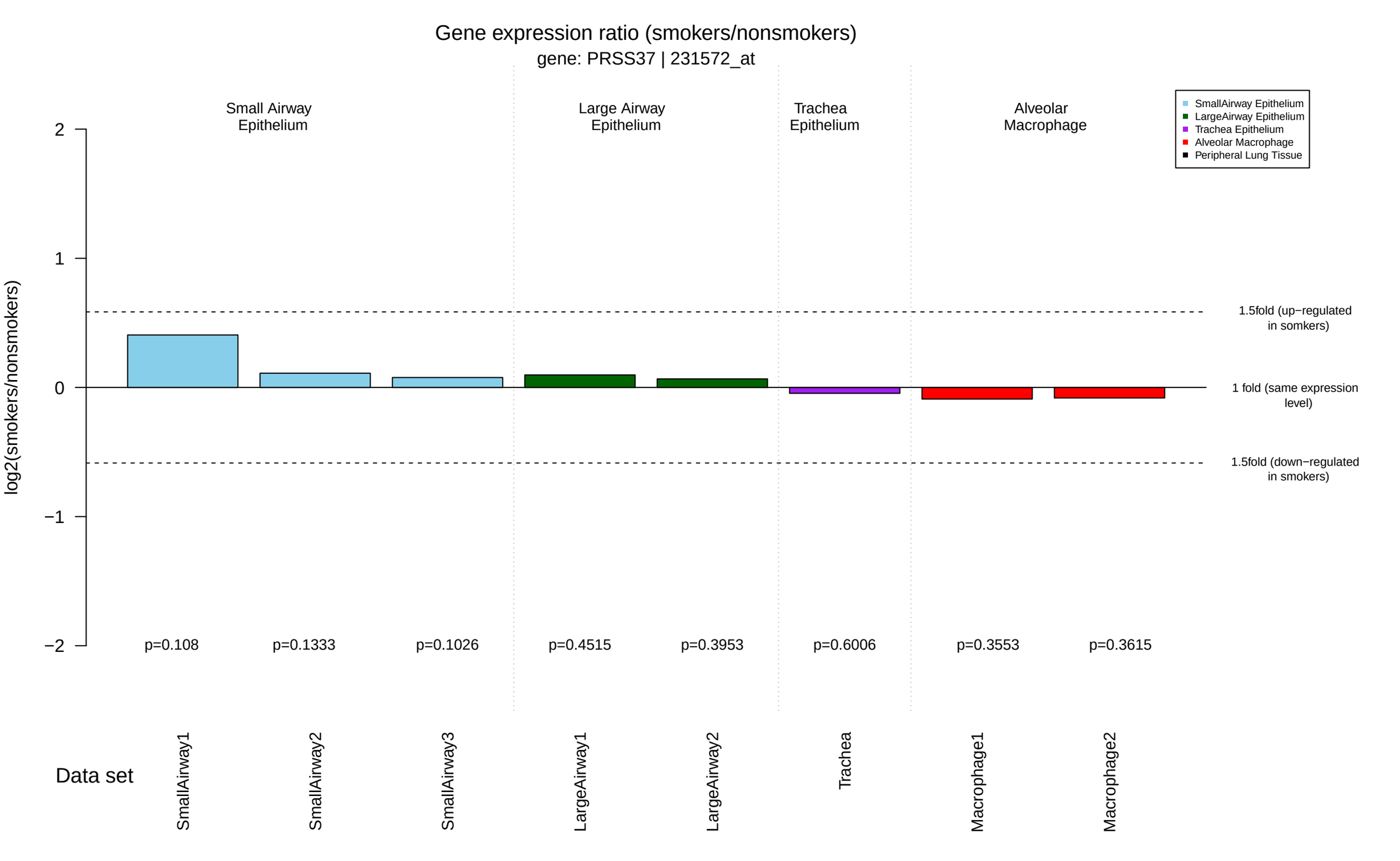
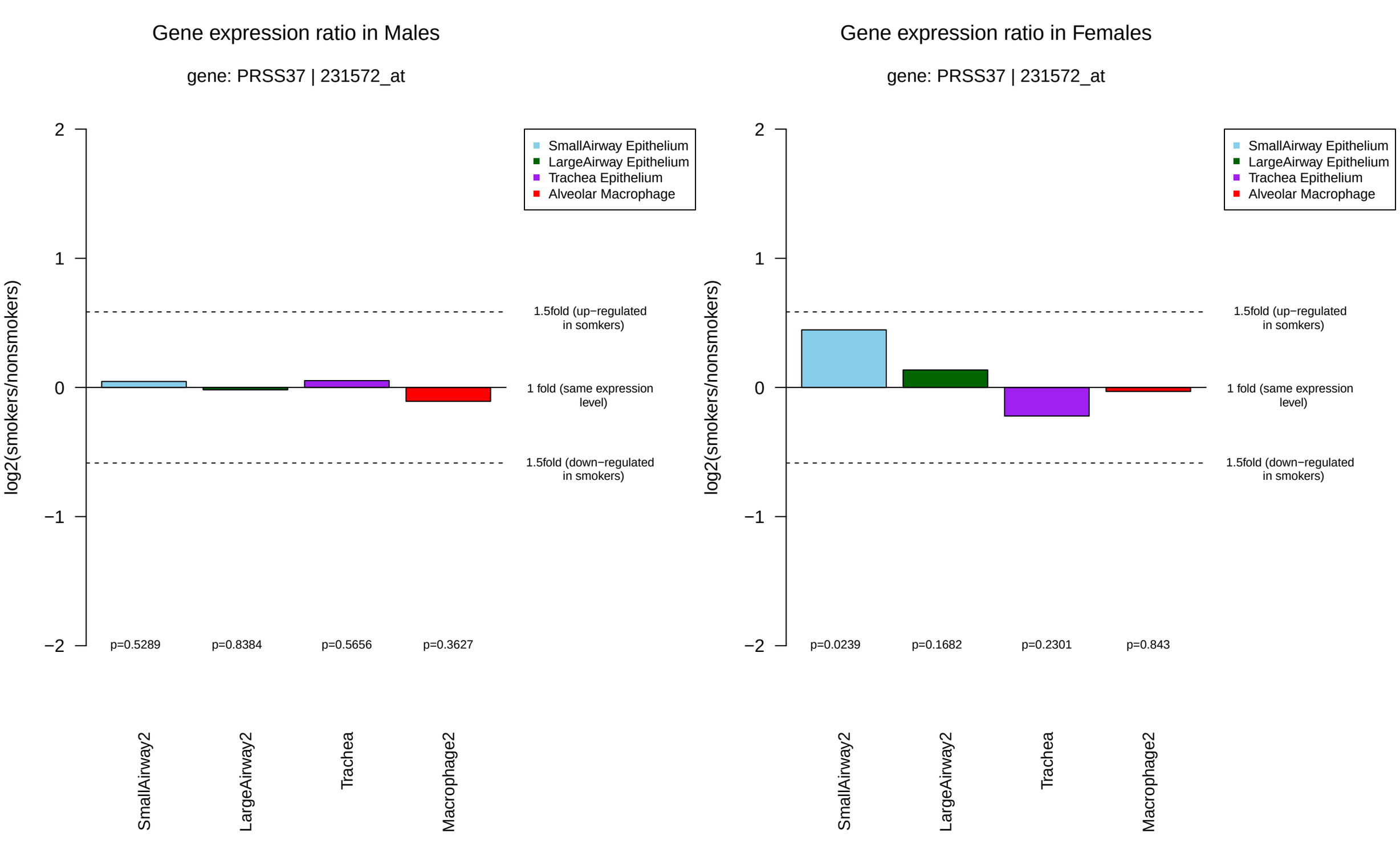
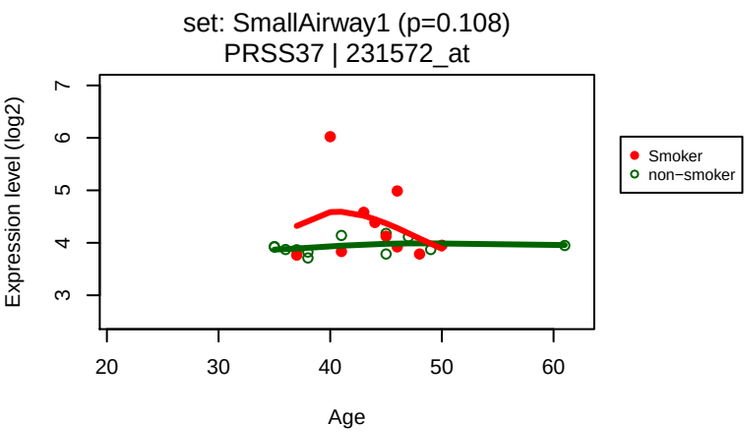
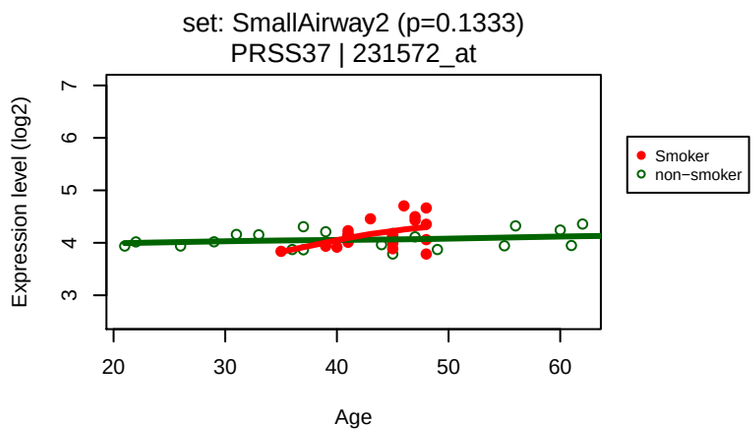
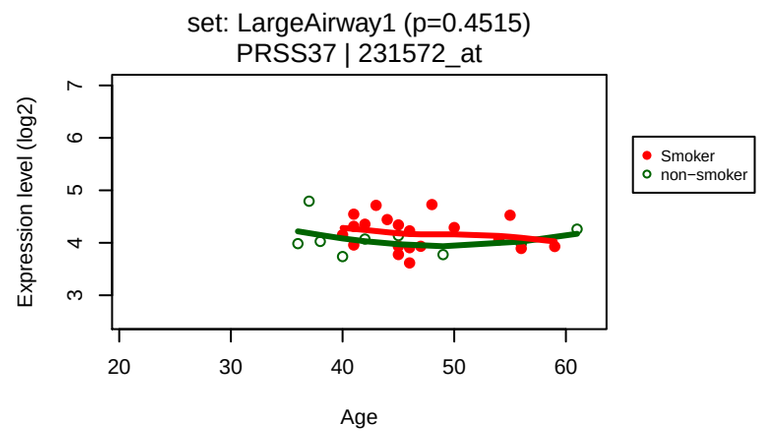

Gene expression of smokers and nonsmokers
gene: PRSS37 | 231572_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.108
p=0.1333
p=0.1026
p=0.4515
p=0.3953
p=0.6006
p=0.3553
p=0.3615
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

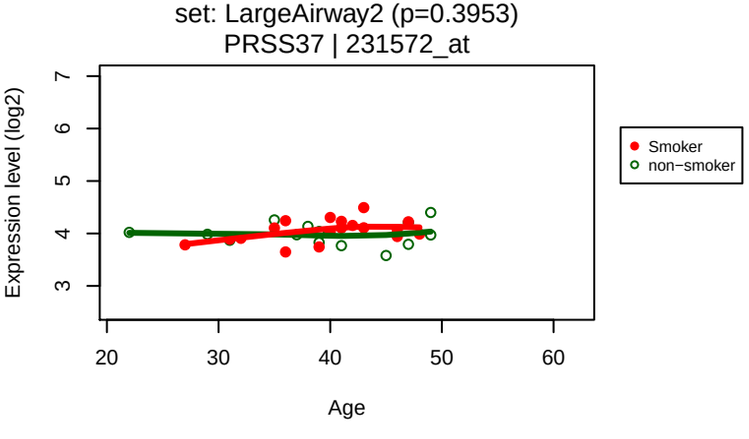

Gene expression ratio (smokers/nonsmokers)
gene: PRSS37 | 231572_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.108
p=0.1333
p=0.1026
p=0.4515
p=0.3953
p=0.6006
p=0.3553
p=0.3615
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

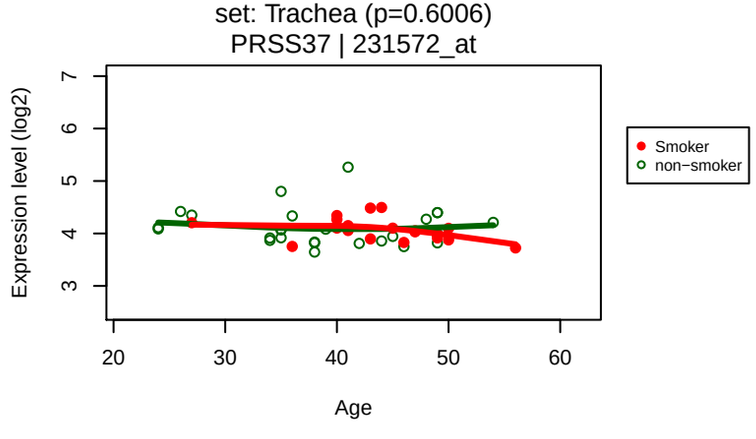

Gene expression ratio in Males
gene: PRSS37 | 231572_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5289
p=0.8384
p=0.5656
p=0.3627
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PRSS37 | 231572_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0239
p=0.1682
p=0.2301
p=0.843
SmallAirway2
LargeAirway2
Trachea
Macrophage2

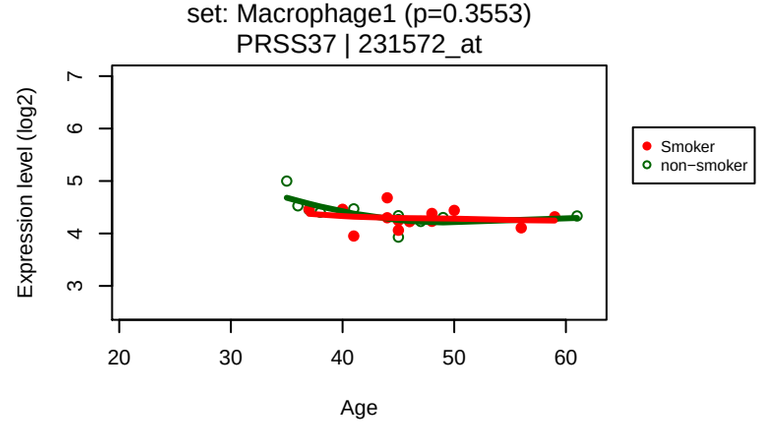

set: SmallAirway1 (p=0.108)
PRSS37 | 231572_at
Expression level (log2)
Age
Smoker
non−smoker

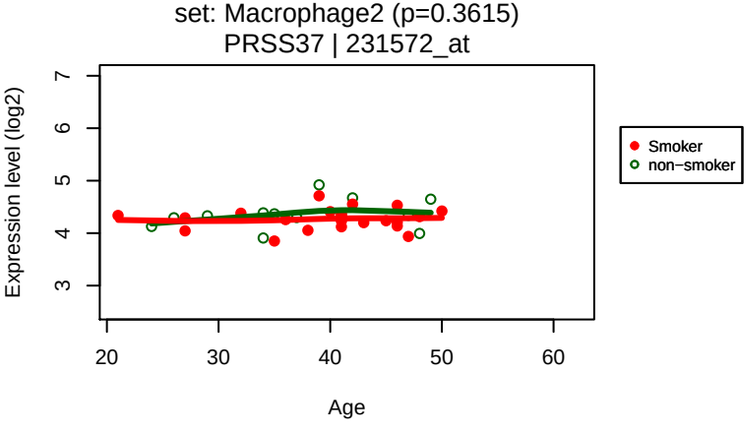

set: SmallAirway2 (p=0.1333)
PRSS37 | 231572_at
Expression level (log2)
Age
Smoker
non−smoker

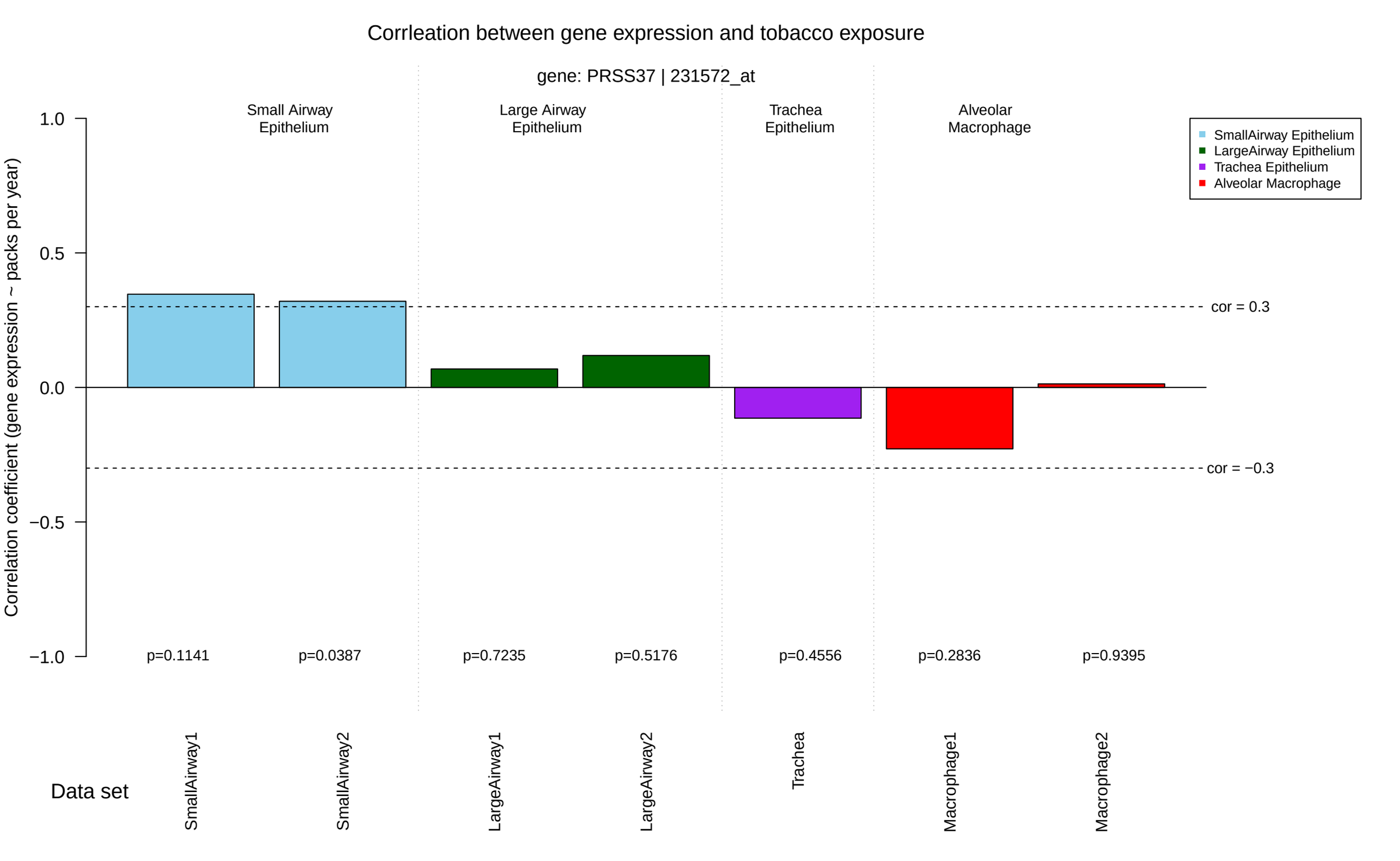

set: LargeAirway1 (p=0.4515)
PRSS37 | 231572_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3953)
PRSS37 | 231572_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6006)
PRSS37 | 231572_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3553)
PRSS37 | 231572_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3615)
PRSS37 | 231572_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PRSS37 | 231572_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1141
p=0.0387
p=0.7235
p=0.5176
p=0.4556
p=0.2836
p=0.9395
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2